Paper I — Identification: $\log\zeta$ as Categorical Effective Action on the Prime-Axis Simplex, and the $\mathbb{Q}(\sqrt{-3})$ Galois Decomposition of the $\mathbb{Z}/6$ Hurwitz Packet

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Abstract

Situation. The trilogy *Identifying 3* — `1` , `2` , `3` — supplied a chart, a cumulant tower, and a character basis on the conductor lattice of $\mathbb{Z}/N$, ending at the cubic selection rule $a+b+c \equiv 0 \pmod N$. The trilogy did not specify which exponential family to read on the chart, or what cubic structure to expect off the uniform distribution.

Task. Identify the natural exponential family for the Riemann zeta function, derive the cubic Amari–Chentsov tensor of its log-partition in closed form, locate the critical line as a structural locus on the primorial ladder, and resolve the smallest squarefree multi-prime modulus ($N = 6$) into its finite-arithmetic Galois content.

Action. Take the prime-axis multi-index space $\mathbb{N}^{(\mathcal{P})}$ with $X_p(n) = k_p(n)$ the p-adic valuation. The categorical exponential family on $\mathbb{N}_{>0}$ with natural parameters $(\eta_p)_p$ has log-partition $\Phi(\eta) = -\sum_p \log(1 - e^{\eta_p})$. Specialize along $\eta_p(s) = -s \log p$. Read the cumulant tower in `3` 's character basis on the residue-projected zeta distribution $Q^{(N)}(s)$. Use Gauss-sum inversion to translate additive to multiplicative selection rules. At $N = 6$, expand the Hurwitz channels in closed form. Identify their 2-dimensional span with the Galois decomposition of the Dedekind zeta of $\mathbb{Q}(\sqrt{-3})$.

Result. The Riemann zeta function $\log\zeta(s)$ is `2` 's categorical effective action on the prime-axis exponential family (**Theorem 2.1**). The cubic Amari–Chentsov tensor of $\log\zeta$ on $\mathbb{Z}/N$ at the residue-projected zeta distribution has the closed-form polynomial expression $T^{\zeta}(\chi_a, \chi_b, \chi_c; s, N) = \zeta(s)^{-1}\zeta_N(s, a+b+c) - \zeta(s)^{-2}\sum_{(ij)} \zeta_N(s, i+j)\zeta_N(s, k) + 2\,\zeta(s)^{-3}\prod_j \zeta_N(s, j)$ in the additive twists $\zeta_N(s, k) = \sum_m e^{2\pi i k m/N}/m^s$ (**Theorem 3.1**); it recovers the trilogy selection rule as its $s \to 1^+$ boundary value. The cubic functional equation $|\hat{Q}_k^{(N)}(s)| = N^{1/2-\mathrm{Re}(s)}\, d_k^{\mathrm{Re}(s)-1/2}\, |\hat{Q}_{-k}^{(N)}(1-s)|$ pins $\mathrm{Re}(s) = 1/2$ as the unique N-uniformity locus along the primorial ladder (**Theorem 4.1**). The five non-trivial Hurwitz channels $\zeta(s, r/6)$ for $r \in \{1, 2, 3, 4, 5\}$ live in a 2-dimensional function space spanned by $\zeta(s)$ and $L(s, \chi_3)$, with explicit closed-form coefficients in $a = 2^{-s}, b = 3^{-s}$ (**Theorem 5.1**); the K-involution $r \to -r \pmod 6$ block-diagonalizes the collapse into a K-fixed ζ-channel and a K-anti-fixed L_{χ_3}-channel (**Theorem 5.2**). This 2-channel structure is, on the nose, the Galois decomposition of the Dedekind zeta $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \zeta(s) \cdot L(s, \chi_3)$ for the degree-2 cover $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$, with K-involution = Galois involution and the program's $\mathbb{Z}/6$ residue chart = combinatorial datum of roots of unity in the Connes–Consani Picard groupoid for this cover (**Identification T3.3**). The program's $\mathbb{Z}/6$ Bures carrier is therefore structurally the degree-2 cover by $\mathbb{Q}(\sqrt{-3})$ of the Connes–Consani adele class space (**Identification I3.4**). Two empirical probes confirm the closed forms to machine precision (Hurwitz collapse: $\sim 10^{-60}$ analytical agreement; Dedekind factorization: seven-digit class-number-formula agreement).

Position. Paper I is the first of three papers in the post-trilogy **RH-Done** sequence: identification (this paper), the right carrier (Paper II — occupation-decorated symmetric Fock + Uniform Place Treatment + CM Poisson Monotonicity), closure (Paper III — convolution-square positivity + averaged CRT shadow uniformity + positive-current transport). Paper I establishes **what the program is working with**; it does not construct the Hilbert space (Paper II) or prove the Riemann Hypothesis (Paper III).

Keywords: Riemann zeta function, information geometry, categorical exponential family, Amari–Chentsov tensor, conductor packet, Dirichlet characters, Hurwitz zeta, Galois decomposition, Connes–Consani adele class space, $\mathbb{Q}(\sqrt{-3})$, prime-axis simplex.

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Introduction

The trilogy *Identifying 3* — `1` [`Murillo2026a`], `2` [`Murillo2026b`], `3` [`Murillo2026c`] — supplied a substrate whose shape does not depend on which way the reader prefers to read it. From information geometry it is the native simplicial chart, the cumulant tower of a single log-partition, and the Pontryagin character basis on $\mathbb{Z}/N$. From discrete harmonic analysis it is the same chart together with the DFT-diagonalization of the categorical Fisher tensors. From classical analytic number theory it is the beginning of a dictionary: at the third cumulant, evaluated at the uniform distribution on $\mathbb{Z}/N$, the single selection rule

$$a + b + c \equiv 0 \pmod{N}$$

is the same line of modular arithmetic that controls diagonal contributions to triple L-function moments. The trilogy stopped at the selection rule. What sits on it has been the program's question ever since.

Paper I makes the answer explicit. The Riemann zeta function $\log \zeta(s)$ is the trilogy's categorical effective action — `2` 's $\Phi = \log Z$ — read on the prime-axis exponential family. The cubic Amari–Chentsov tensor of $\log \zeta$ in `3` 's residue chart on $\mathbb{Z}/N$ has a closed-form expression, graded by the divisor lattice of N, that recovers the trilogy selection rule as its high-temperature boundary value. The cubic functional equation that this tensor satisfies pins the critical line $\mathrm{Re}(s) = 1/2$ as the unique locus on which the moduli of its twisted Plancherel coefficients are uniformly bounded in N along the primorial ladder. At the smallest squarefree multi-prime modulus, $N = 6$, the five non-trivial Hurwitz channels $\zeta(s, r/6)$ for $r \in \{1, 2, 3, 4, 5\}$ live not in a 5-dimensional space but in the 2-dimensional space spanned by $\zeta(s)$ and the unique primitive Dirichlet L-function modulo 3, $L(s, \chi_3)$. This 2-channel system is precisely the Galois decomposition of the Dedekind zeta of $\mathbb{Q}(\sqrt{-3})$ under the degree-2 cover $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$: the program's K-involution on the $\mathbb{Z}/6$ Bures fiber is the Galois involution $\sqrt{-3} \mapsto -\sqrt{-3}$.

These are the **identifications** Paper I delivers. They establish *what we are working with* — they do not yet build the right Hilbert space (Paper II) or close to a proof of the Riemann Hypothesis (Paper III).

Position in the program

The trilogy `Identifying 3` is published [`Murillo2026a`, `Murillo2026b`, `Murillo2026c`]. The present paper is the first of a second three-paper sequence, **RH-Done**, of which the planned components are:

- **Paper I — Identification** (this paper). What the program is working with. Dots 1–3 of the spine [`RH-Done-Draft1`, §A].
- **Paper II — The Right Carrier**. The occupation-decorated symmetric Fock that hosts an exact von Mangoldt formula upstairs of the CRT compression; the Uniform Place Treatment

that promotes this finite carrier to the Archimedean place by Stone–von Neumann uniqueness; the CM Poisson Monotonicity criterion. Dots 4–6.

- **Paper III — Closure**. Convolution-square positivity of Poisson windows, averaged CRT shadow uniformity at polynomial scale, the positive-current transport theorem, and the main theorem on RH via Polya–Laguerre. Dots 7–10.

Paper I sits at the seam between two trilogies. It cites `,1 ,2 3` as established foundation; it sets the stage for `Paper II` by identifying the right finite arithmetic substrate.

What is new here, what is cited

The load-bearing identifications of Paper I were sketched in working notes circulated within the program; we re-derive them in clean form here. The provenance maps section by section as follows.

- **Theorem T2.1** *(Prime-Axis Effective Action; §2.5)* — clean rewrite of *siete* §2's substitution $\eta_p(s) = -s\log p \Rightarrow \Phi(\eta(s)) = \log\zeta(s)$, integrated with the trilogy notation. No new analytic content; the partition-function-as-Euler-product identification is the same as in *siete*.
- **Theorem T2.2** *(Closed-Form Cubic A-C Tensor on $\mathbb{Z}/N$; §3.4)* — clean rewrite of *siete* §5's closed-form polynomial in $\zeta_N(s,k)/\zeta(s)$. The explicit term-by-term contraction of $T_{ijk}(Q^{(N)})$ against $\chi_a \otimes \chi_b \otimes \chi_c$ in the proof of Theorem 3.1 is new in Paper I (the *siete* derivation jumps through one intermediate step).
- **Theorem T2.3** *(Cubic Functional Equation and Critical-Line N-Uniformity; §4.5)* — clean rewrite of *siete* §7's modulus identity $|\widehat{Q}_k^{(N)}(s)| = N^{1/2-\mathrm{Re}(s)}\, d_k^{\mathrm{Re}(s)-1/2}\, |\widehat{Q}_{-k}^{(N)}(1-s)|$. The three-"1/2"s compatibility paragraph (§4.6), reconciling §4.5's critical-line identification with the Riemann, Mellin, and trilogy 1/2's, is new in Paper I.
- **Theorem T3.1** *(Closed Form for $\zeta(s, r/6)$; §5.2)* — promotion of [HPo-notes, Addendum A1] to a formal theorem with a three-step classical proof (half-shift collapse, mod-3 split, joint mod-6 reduction). New in Paper I.
- **Theorem T3.2** *(K-Orbit Parity Decomposition; §5.3)* — promotion of [HPo-notes, Addendum A2–A3] to a formal theorem. New in Paper I.
- **Identification T3.3** *(Galois Identification with $\mathbb{Q}(\sqrt{-3})$; §6.4)* — promotion of [HPo-notes, Addendum H1–H3] to an entry-by-entry identification table, with row-by-row justification. New in Paper I.
- **Identification I3.4** *(The $\mathbb{Z}/6$ Bures carrier as the degree-2 cover of the Connes–Consani adele class space; §6.5)* — closing identification combining T3.3 with the program's carrier language. New in Paper I.
- **Theorem T5.3** *(Unconditional non-vanishing of $L(1/2, \chi_3)$; §5.6)* — synthesis of Iwaniec–Sarnak (2000) non-negativity plus the LMFDB record `1-3-3.2-r1-0-0`, recorded in [HPo-notes, Addendum A7]. The formal statement is new in Paper I; the result itself is classical via the cited sources.

The numerical verifications underlying T3.1 and T3.3 are recorded in the probes [`probe_Hurwitz_z6_closed_form.` (verifies T3.1 + T3.2 to 60-digit `mpmath` precision; verifies K-even zero-ordinate constraints to machine epsilon) and [`probe_DedekindFactorization_Qsqrt-3_ClaudeEinstein.py`] (verifies Dedekind factorization against finite Euler product on norms $\leq$ 200; class-number-formula agreement to $\sim 10^{-7}$).

Two audiences

The paper is written for two readerships, and bridges between them.

For the information geometer. The conjugacy between η-coordinates and p-coordinates, the cumulant tower of $\Phi = \log Z$, the Amari–Chentsov tensor, the conductor packet H_d, and the character selection rule on $\mathbb{Z}/N$ are all assumed familiar from the trilogy. The analytic-number-theoretic objects entering Paper I — Dirichlet L-functions $L(s,\chi)$, the Hurwitz zeta $\zeta(s,r/N)$, the additive twist $\zeta_N(s,k)$, Gauss sums, and the functional equation of ζ — are introduced with one-line glosses where they appear, but without re-derivation; the standard reference is [Iwaniec & Kowalski, 2004].

For the analytic number theorist. ζ, Dirichlet characters, and the explicit-formula bridge are assumed familiar. The information-geometric objects entering Paper I — the categorical exponential family on H_0, the Legendre duality $\eta \leftrightarrow p = \nabla\Phi$, the Fisher information Σ as a covariance, the Amari–Chentsov tensor T_{ijk} as a third cumulant, and the conductor packet H_d on $\mathbb{Z}/N$ — are introduced with one-line glosses where they appear; the standard reference is [Amari & Nagaoka, 2000]. The reader who is willing to import a chart in which $\log\zeta$ is the natural object will find Paper I a chain of identifications, each one classical in either of the two languages.

Roadmap

§1 recaps the trilogy in three pages. §§2–4 execute the *siete* identifications (T2.1, T2.2, T2.3) with full derivations. §5 promotes the $\mathbb{Z}/6$ 2D collapse from the HPo Addendum A working note into a clean theorem (T3.1, T3.2). §6 carries that into the Galois identification with $\mathbb{Q}(\sqrt{-3})$ (T3.3, I3.4) following HPo Addendum H. §7 closes with the hand-off to Paper II.

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§1. The Trilogy in One Page

This section is cite-and-use. It collects the three theorems and one identification from `Identifying 3` that Paper I needs as input, in the form in which they are quoted later. No proofs are given here; all are in the published trilogy.

1.1 The native chart (`1`)

The probability simplex $\Delta^{N-1} \subset \mathbb{R}^N$ over N outcomes has its natural tangent space at any interior point given by the zero-sum hyperplane

$$H_0 \;=\; \{x \in \mathbb{R}^N \,:\, \textstyle\sum_i x_i = 0\}.$$

Through the square-root map $\psi : \Delta^{N-1} \to S^{N-1}_+$, $\psi(p) = (\sqrt{p_1}, \dots, \sqrt{p_N})$, the Fisher–Rao metric g_{FR} on Δ^{N-1} is the pullback of $4g_S$, where g_S is the round sphere metric on S^{N-1}_+. At the uniform distribution $u = \frac{1}{N}(1, \dots, 1)$ the Fisher–Rao isometry group is

$$\mathrm{Stab}_{g_{\mathrm{FR}}}(u) \;=\; \mathrm{SO}(H_0) \;\cong\; \mathrm{SO}(N-1).$$

This is **Theorem T1.1**.

1.2 The cumulant tower (`2`)

Take $\eta = (\eta_1, \dots, \eta_N) \in H_0$ as natural parameters of the categorical exponential family

$$p_i(\eta) \;=\; \frac{e^{\eta_i}}{Z(\eta)}, \qquad Z(\eta) \;=\; \sum_{j=1}^{N} e^{\eta_j},$$

and define the **categorical effective action**

$$\boxed{\Phi(\eta) \;:=\; \log Z(\eta).}$$

Successive derivatives of Φ in η generate the cumulant tower of the sufficient statistic X. The first four orders are

$$\partial_i\Phi \;=\; p_i \quad \text{(mean / position)}, \qquad \partial_i\partial_j\Phi \;=\; p_i\delta_{ij} - p_ip_j \quad \text{(Fisher information } \Sigma_{ij}\text{)},$$

$$\partial_i\partial_j\partial_k\Phi \;=\; p_i\delta_{ij}\delta_{ik} - p_ip_k\delta_{ij} - p_ip_j\delta_{ik} - p_ip_j\delta_{jk} + 2p_ip_jp_k \quad \text{(Amari–Chentsov tensor } T_{ijk}\text{)},$$

$$\partial_i\partial_j\partial_k\partial_\ell\Phi \;=\; Q_{ijk\ell} \quad \text{(closure defect / fourth cumulant)}.$$

One scalar functional Φ generates four named geometric objects through plain differentiation. This is **Theorem T1.2**. The leap from order 2 (Riemannian, Gaussian-visible) to order 3 (Amari–Chentsov, the first invariant beyond Riemannian geometry) is the trilogy's titular *Identifying 3*; we will write **I1.4** for this structural observation when we need to label it.

1.3 The character selection rule (3)

Identify the outcome index set with the cyclic ring $\mathbb{Z}/N$. The Pontryagin character basis on $\mathbb{C}^N$ is

$$\chi_k(x) \;=\; e^{2\pi ikx/N}, \qquad k, x \in \mathbb{Z}/N.$$

The non-trivial characters $\{\chi_k\}_{k=1,\dots,N-1}$ span the complexification $H_0 \otimes \mathbb{C}$. Each χ_k has a *conductor* $d_k := N/\gcd(k,N)$, equal to the smallest divisor of N on which χ_k is natively defined. The conductor packet

$$H_d \;:=\; \operatorname{span}_{\mathbb{C}}\{\chi_k \,:\, d_k = d\}, \qquad \dim_{\mathbb{C}} H_d \;=\; \varphi(d),$$

gives the decomposition

$$H_0 \otimes \mathbb{C} \;=\; \bigoplus_{\substack{d \mid N \\ d>1}} H_d.$$

Read 2 's tensors at the uniform u in this basis. The Fisher information becomes diagonal — in fact scalar:

$$\Sigma(\chi_a, \chi_b)\big|_u \;=\; \delta_{a,b} \quad \text{for } a, b \in \{1, \dots, N-1\}.$$

The Amari–Chentsov tensor satisfies the **selection rule**

$$\boxed{T(\chi_a, \chi_b, \chi_c)\big|_u \;=\; \delta_{a+b+c \equiv 0 \pmod N}, \quad a, b, c \in \{1, \dots, N-1\}.}$$

This is **Theorem T1.3**. The second cumulant is silent on arithmetic; the third cumulant carries the modular constraint.

1.4 What Paper I adds

The trilogy stopped at the selection rule. It established that **if** the categorical exponential family of 2 is read on a $\mathbb{Z}/N$-structured outcome set, **then** the third cumulant at the uniform satisfies $a + b + c \equiv 0 \pmod N$. It did not specify *which* exponential family one should care about, or *what* the resulting cubic structure looks like *off* the uniform.

The present paper specifies. The right exponential family — for the Riemann Hypothesis — is the prime-axis family of §2 below, with $\Phi = \log\zeta$. Its cubic Amari–Chentsov tensor at the residue-projected zeta distribution on $\mathbb{Z}/N$ has a closed form (T2.2). Its functional equation pins the

critical line (T2.3). At $N = 6$ the five non-trivial Hurwitz channels collapse to a 2D system (T3.1, T3.2). And the 2D system is *exactly* the Galois decomposition of $\zeta_{\mathbb{Q}(\sqrt{-3})}$ (T3.3, I3.4). The trilogy supplied the substrate; the present paper names the natural object on it.

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§2. The Prime-Axis Exponential Family and $\log\zeta$

We introduce the prime-axis exponential family on the discrete probability space $\mathbb{N}_{>0}$ and identify its categorical effective action Φ with $\log\zeta(s)$. The identification (Theorem T2.1) is by substitution, not analogy.

2.1 Prime-axis coordinates on $\mathbb{N}_{>0}$

Every positive integer n has a unique factorization

$$n \;=\; \prod_{p\in\mathcal{P}} p^{k_p(n)}, \qquad k_p(n)\in\mathbb{N},$$

where $\mathcal{P} = \{2,3,5,7,11,...\}$ is the set of primes and only finitely many $k_p(n)$ are nonzero. The map $n \mapsto k(n) = (k_p(n))_p$ is a bijection $\mathbb{N}_{>0} \cong \mathbb{N}^{(\mathcal{P})}$ onto the finitely-supported $\mathbb{N}$-valued sequences indexed by primes. We treat $\mathbb{N}^{(\mathcal{P})}$ as the **prime-axis multi-index space** on which the prime-axis exponential family is defined.

The *sufficient statistic on the p-axis* is $X_p(n) := k_p(n)$, the p-adic valuation of n. The p-axes are statistically independent under the factorization (formalized below).

2.2 The exponential family and its log-partition

Let $\eta = (\eta_p)_{p\in\mathcal{P}}$ be a sequence of complex natural parameters indexed by primes, with $\mathrm{Re}(\eta_p) < 0$ for every p and the regularity condition $\sum_p |e^{\eta_p}|/(1-|e^{\eta_p}|) < \infty$ (which suffices for absolute convergence of every cumulant computed below).

Definition 2.1. The *prime-axis exponential family* on $\mathbb{N}_{>0}$ is the family of probability distributions

$$p_n(\eta) \;=\; \frac{1}{Z(\eta)}\exp\Big(\sum_p k_p(n)\,\eta_p\Big), \qquad Z(\eta) \;=\; \sum_{n=1}^{\infty}\exp\Big(\sum_p k_p(n)\,\eta_p\Big).$$

The partition function factorizes by prime axis. Set $u_p := e^{\eta_p}$; then

$$Z(\eta) \;=\; \sum_n \prod_p u_p^{k_p(n)} \;=\; \prod_p \sum_{k=0}^{\infty} u_p^k \;=\; \prod_p \frac{1}{1-u_p},$$

where the double interchange of sum and product is justified by absolute convergence under our regularity condition. The log-partition is therefore

$$\boxed{\Phi(\eta) \;=\; \log Z(\eta) \;=\; -\sum_p \log(1-e^{\eta_p}).}$$

The Euler product is categorical independence. The factorization of $Z(\eta)$ by prime axis is the partition-function fingerprint of the multiplicative independence of the prime-axis sufficient

statistics $(X_p)_p$. Equivalently: under any η, the joint distribution of $(X_p)_p$ under the prime-axis exponential family is the product of geometric distributions, one per prime, with p-axis parameter $u_p = e^{\eta_p}$:

$$\Pr_\eta(X_p = k) \;=\; (1-u_p)\,u_p^k, \qquad k \in \mathbb{N}.$$

The cumulant tower of Φ is, by construction, **diagonal in primes at every order**: $\partial_{\eta_p}\partial_{\eta_q}\Phi = 0$ for $p \neq q$, and similarly for all higher mixed partials.

2.3 Specialization to $\eta_p(s) = -s\log p$

Take a one-parameter family of natural parameters along the *diagonal slice*

$$\eta_p(s) \;:=\; -s\log p, \qquad p \in \mathcal{P}, \quad s \in \mathbb{C}.$$

For $\mathrm{Re}(s) > 1$ the regularity condition of §2.2 is satisfied (this is the standard convergence region of ζ), and $e^{\eta_p(s)} = p^{-s}$. Substituting into the log-partition,

$$\Phi(\eta(s)) \;=\; -\sum_p \log(1-p^{-s}) \;=\; \log\zeta(s).$$

We write $\Phi_\zeta(s) := \log\zeta(s)$ for this *diagonal specialization* of Φ, and we analytically continue Φ_ζ to the standard domain $\mathbb{C}\setminus\{1\}$ in the usual way (from this point onward Φ_ζ means the analytically continued object).

The zeta distribution. Under the natural parameter $\eta(s)$ at real $s > 1$, the prime-axis exponential family on $\mathbb{N}_{>0}$ has mean-parameter distribution

$$q_n(s) \;=\; \prod_p (1-p^{-s})\,p^{-sk_p(n)} \;=\; \frac{1}{\zeta(s)}\,n^{-s},$$

which is the classical **Riemann zeta distribution** on $\mathbb{N}_{>0}$.

2.4 The Mellin/amplitude reading

Through the trilogy's square-root map ψ (§1.1) applied to $q(s)$, the amplitude vector on $\mathbb{N}_{>0}$ is

$$a_n(s) \;=\; \sqrt{q_n(s)} \;=\; \frac{n^{-s/2}}{\sqrt{\zeta(s)}}.$$

The Bhattacharyya coefficient of two zeta amplitudes at parameters s_1, s_2 is

$$\langle a(s_1), a(s_2)\rangle \;=\; \sum_n a_n(s_1)\,a_n(s_2) \;=\; \frac{\zeta((s_1+s_2)/2)}{\sqrt{\zeta(s_1)\,\zeta(s_2)}}.$$

The Bhattacharyya pairing is ζ-valued at the midpoint $(s_1+s_2)/2$. Taking $s_1 = 1/2+it$, $s_2 = 1/2-it$ (a functional-equation conjugate pair on the critical line) gives midpoint $1/2$, so

$$\langle a(1/2+it), a(1/2-it)\rangle \;=\; \frac{\zeta(1/2)}{|\zeta(1/2+it)|}.$$

The central value $\zeta(1/2)$ is the Bhattacharyya inner product of the amplitude vector of the zeta distribution with its functional-equation conjugate. We note this without claiming it is a new analytic fact; the content is structural — $\zeta(1/2)$ is doing geometric work in the trilogy's chart that the chart was not designed to host.

2.5 Theorem T2.1: Prime-Axis Effective Action

We now state the identification.

Theorem 2.1 (*Prime-Axis Effective Action; T2.1 of [RH-Done-Draft1]*). *The Riemann zeta function* $\log\zeta(s)$ *is the categorical effective action* $\Phi(\eta(s))$ *of the prime-axis exponential family of §2.2, evaluated along the diagonal slice* $\eta_p(s) = -s\log p$*:*

$$\Phi_\zeta(s) \;=\; \Phi(\eta(s)) \;=\; -\sum_p \log(1-p^{-s}) \;=\; \log\zeta(s).$$

The corresponding mean-parameter distribution is the Riemann zeta distribution $q_n(s) = n^{-s}/\zeta(s)$ *on* $\mathbb{N}_{>0}$*.*

Proof. The factorization argument of §2.2 gives $\Phi(\eta) = -\sum_p \log(1-e^{\eta_p})$. The substitution $\eta_p(s) = -s\log p$ of §2.3 specializes this to $-\sum_p \log(1-p^{-s})$, which is the standard Euler product representation of $\log\zeta(s)$ on $\mathrm{Re}(s) > 1$. Analytic continuation extends the identification to $s \in \mathbb{C}\setminus\{1\}$. The mean-parameter identification of $q_n(s)$ with the Riemann zeta distribution is direct substitution. □

2.6 The cumulant tower of Φ in the prime-axis chart

Because Φ is additive over primes, every cumulant tensor of Φ in the prime-axis chart is diagonal in primes. On a single prime axis, set $u = e^{\eta_p}$ and write $\Phi^{(p)}(\eta_p) = -\log(1-u)$. Repeated application of the chain rule $d/d\eta_p = u\cdot d/du$ yields

$$\partial^n_{\eta_p}\Phi^{(p)} \;=\; \mathrm{Li}_{-(n-1)}(u) \;=\; \sum_{k=1}^{\infty} k^{n-1}\,u^k, \qquad n \geq 1,$$

the *negative-index polylogarithm*. Equivalently, with $A_{n-1}(u)$ the $(n-1)$st Eulerian polynomial,

$$\partial^n_{\eta_p}\Phi^{(p)} \;=\; \frac{u\,A_{n-1}(u)}{(1-u)^n}.$$

The successive geometric-distribution cumulants on the p-axis emerge as the values of these polylogarithms at $u = e^{\eta_p}$.

Specializing to $\eta_p = -s\log p$ gives the per-axis cumulants

$$\Sigma_{pp}(s) \;=\; \frac{p^{-s}}{(1-p^{-s})^2}, \qquad T_{ppp}(s) \;=\; \frac{p^{-s}(1+p^{-s})}{(1-p^{-s})^3}, \qquad \ldots,$$

with all cross-prime entries vanishing. **The prime-axis chart is silent on arithmetic.** All cubic content is diagonal; no cross-prime selection rule is visible.

This is not a defect; it is a constraint on what charts can see. Arithmetic content lives off the prime-axis chart, in dual coordinates. The $\mathbb{Z}/N$ residue chart of §3 is the natural dual: there the character selection rule of T1.3 becomes the operational support condition.

> The prime-axis chart and the residue chart are Fourier duals. The prime-axis chart trivializes the cumulant tower (diagonal in primes); the residue chart trivializes the arithmetic structure (selection rule on character indices). The cubic A-C tensor of Φ_ζ exhibits both faces — as a sum of single-axis polylogs in the prime-axis chart, and as a finite linear combination of Dirichlet L-functions of cyclic moduli in the residue chart.

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§3. The Residue Chart and the Closed-Form Cubic A-C Tensor

We now read $\Phi_\zeta = \log\zeta$ in the trilogy's character basis on $\mathbb{Z}/N$. Section 3.1 defines the residue-projected zeta distribution $Q^{(N)}(s)$; section 3.2 introduces the *additive-twist zeta* $\zeta_N(s,k)$ as the natural Fourier coefficient of $Q^{(N)}(s)$; section 3.3 recalls the conductor-packet decomposition for primorial N; section 3.4 derives the **closed-form formula** for the cubic A-C tensor on the zeta distribution; section 3.5 verifies that this formula recovers T1.3's selection rule in the $s \to 1^+$ high-temperature limit; section 3.6 records the Gauss-sum duality identifying additive-character selection with multiplicative-character orthogonality of triple L-moments.

3.1 The residue-projected zeta distribution

Fix $N \geq 2$. The reduction-mod-N map $r_N : \mathbb{N}_{>0} \to \mathbb{Z}/N$, $r_N(m) = m \bmod N$, pushes the zeta distribution of §2.3 to a discrete distribution on $\mathbb{Z}/N$.

Definition 3.1. For $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$ and $r \in \mathbb{Z}/N$,

$$Q_r^{(N)}(s) \;:=\; \Pr_{X\sim q(s)}[X \equiv r \pmod N] \;=\; \frac{1}{\zeta(s)} \sum_{m\equiv r \pmod N} m^{-s}.$$

For real $s > 1$, $Q_r^{(N)}(s) > 0$ for every r and $\sum_r Q_r^{(N)}(s) = 1$, so $Q^{(N)}(s) \in \Delta^{N-1}$. For complex s with $\mathrm{Re}(s) > 1$, $Q^{(N)}(s)$ is a normalized signed measure on $\mathbb{Z}/N$; for $\mathrm{Re}(s) \leq 1$ both $\zeta(s)$ and the inner sum acquire poles and the normalization becomes problematic. The analytic identifications below are stated on $\mathrm{Re}(s) > 1$ and extended by meromorphic continuation in the usual way.

3.2 The additive-twist zeta and the Fourier coefficients of $Q^{(N)}$

Read $Q^{(N)}(s)$ in T1.3's character basis $\{\chi_k(r) = e^{2\pi ikr/N}\}_{k\in\mathbb{Z}/N}$:

$$\widehat{Q}_k^{(N)}(s) \;:=\; \sum_{r\in\mathbb{Z}/N} Q_r^{(N)}(s)\,\chi_k(r) \;=\; \frac{1}{\zeta(s)} \sum_{m=1}^{\infty} \frac{e^{2\pi ikm/N}}{m^s}.$$

Definition 3.2. The *additive-twist zeta* at modulus N and frequency k is

$$\boxed{\zeta_N(s,k) \;:=\; \sum_{m=1}^{\infty} \frac{e^{2\pi ikm/N}}{m^s}.}$$

Then

$$\widehat{Q}_k^{(N)}(s) \;=\; \frac{\zeta_N(s,k)}{\zeta(s)}.$$

Two boundary values are immediate.

- **Trivial frequency $k = 0$:** $\zeta_N(s,0) = \zeta(s)$, so $\widehat{Q}_0^{(N)}(s) = 1$ — the probability normalization.
- **High-temperature limit $s \to 1^+$:** by the Wiener–Ikehara theorem applied to $\sum_{m\equiv r \pmod N} m^{-s}$ — equivalently, by the primitive-character L-function dictionary of §3.6 below combined with the simple pole structure of $L(s,\chi_0) = \zeta(s)\prod_{p\mid N}(1-p^{-s})$ — every $Q_r^{(N)}(s) \to 1/N$ as $s \to 1^+$. Equivalently, $\widehat{Q}_k^{(N)}(s) \to \delta_{k\equiv 0 \pmod N}$.

The second observation says that the residue-projected zeta distribution *continuously deforms* to the uniform distribution on $\mathbb{Z}/N$ as $s \to 1^+$. This is the natural base point for reading the trilogy's selection rule.

3.3 Conductor packets on $\mathbb{Z}/N$

We recall T1.3's conductor-packet decomposition (3 §3) and specialize to the case $N = P_n := p_1 p_2 \cdots p_n$ of a primorial modulus, which is the natural ladder along which Paper II's transport theorem will compute.

Each character χ_k on $\mathbb{Z}/N$ has conductor $d_k := N/\gcd(k, N)$. For squarefree N — in particular $N = P_n$ — every divisor $d \mid N$ corresponds to a subset $S \subseteq \{p_1, \dots, p_n\}$ with $d = \prod_{p \in S} p$. The conductor packet

$$H_d \;=\; \operatorname{span}_{\mathbb{C}}\{\chi_k \,:\, d_k = d\}$$

has dimension $\varphi(d) = \prod_{p \in S}(p-1)$. The total dimension of $H_0(\mathbb{Z}/P_n)$ is $P_n - 1 = \sum_{d \mid P_n,\, d>1} \varphi(d)$, which is a special case of $\sum_{d \mid N} \varphi(d) = N$.

The Fourier coefficient $\widehat{Q}_k^{(N)}(s) = \zeta_N(s,k)/\zeta(s)$ depends on k only through its conductor d_k (up to the finite-character linear combination of §3.6 below). It therefore **lives in the conductor packet** H_{d_k} of T1.3.

The residue chart on $\mathbb{Z}/P_n$ is the natural setting in which the prime-axis cross-prime structure becomes visible, graded by the divisor lattice of P_n.

3.4 Closed form for the cubic Amari–Chentsov tensor at $Q^{(N)}(s)$

We compute T1.2's third cumulant tensor T_{ijk} at the distribution $p = Q^{(N)}(s)$, read in T1.3's character basis.

From T1.2's formula (substituting $Q_i^{(N)}(s)$ for p_i):

$$T_{ijk}(Q^{(N)}(s)) \;=\; Q_i^{(N)}\,\delta_{ij}\delta_{ik} - Q_i^{(N)}Q_k^{(N)}\delta_{ij} - Q_i^{(N)}Q_j^{(N)}\delta_{ik} - Q_i^{(N)}Q_j^{(N)}\delta_{jk} + 2\,Q_i^{(N)}Q_j^{(N)}Q_k^{(N)}.$$

Contract this with three character vectors χ_a, χ_b, χ_c in the bilinear convention (T1.3 §5), using the Fourier coefficient $\widehat{Q}_k^{(N)}(s) = \sum_r Q_r^{(N)}\,\chi_k(r) = \zeta_N(s,k)/\zeta(s)$ from §3.2. The term-by-term computation is recorded in the proof of Theorem 3.1 below; the resulting closed form is:

Theorem 3.1 (*Closed-Form Cubic A-C Tensor on $\mathbb{Z}/N$; T2.2 of [RH-Done-Draft1]*). *At the residue-projected zeta distribution $Q^{(N)}(s)$, the Amari–Chentsov tensor of $\Phi_\zeta = \log\zeta$ read in T1.3's character basis on $\mathbb{Z}/N$ has closed form*

$$\begin{aligned} T^{\zeta}(\chi_a, \chi_b, \chi_c;\, s, N) \;=\; & \frac{\zeta_N(s, a+b+c)}{\zeta(s)} \\ & - \frac{\zeta_N(s, a+b)\,\zeta_N(s,c) + \zeta_N(s, a+c)\,\zeta_N(s,b) + \zeta_N(s, b+c)\,\zeta_N(s,a)}{\zeta(s)^2} \\ & + \frac{2\,\zeta_N(s,a)\,\zeta_N(s,b)\,\zeta_N(s,c)}{\zeta(s)^3}, \end{aligned}$$

for $a, b, c \in \mathbb{Z}/N \setminus \{0\}$ and $\operatorname{Re}(s) > 1$, with meromorphic continuation in s.

Proof. The contraction of $T_{ijk}(Q^{(N)}(s))$ with $\chi_a \otimes \chi_b \otimes \chi_c$ is the sum

$$\sum_{i,j,k \in \mathbb{Z}/N} T_{ijk}(Q^{(N)}(s))\, \chi_a(i)\, \chi_b(j)\, \chi_c(k).$$

Read each of the five terms of T_{ijk} separately. The Fourier coefficient $\widehat{Q}_k^{(N)}(s) = \sum_r Q_r^{(N)}(s)\, \chi_k(r)$ of §3.2 and the multiplicative property $\chi_a(r)\, \chi_b(r) = \chi_{a+b}(r)$ of additive characters on $\mathbb{Z}/N$ are the only inputs.

Term 1 (linear in Q, double Kronecker $\delta_{ij}\delta_{ik}$). Forcing $i = j = k$:

$$\sum_{i,j,k} Q_i^{(N)}\, \delta_{ij}\delta_{ik}\, \chi_a(i)\chi_b(j)\chi_c(k) \;=\; \sum_i Q_i^{(N)}\, \chi_a(i)\, \chi_b(i)\, \chi_c(i) \;=\; \sum_i Q_i^{(N)}\, \chi_{a+b+c}(i) \;=\; \widehat{Q}_{a+b+c}^{(N)}(s).$$

Term 2 (bilinear in Q, single δ_{ij}). Forcing $i = j$, k free, and factoring:

$$-\sum_{i,j,k} Q_i^{(N)} Q_k^{(N)}\, \delta_{ij}\, \chi_a(i)\chi_b(j)\chi_c(k) \;=\; -\Big(\sum_i Q_i^{(N)}\, \chi_{a+b}(i)\Big)\Big(\sum_k Q_k^{(N)}\, \chi_c(k)\Big) \;=\; -\widehat{Q}_{a+b}^{(N)}(s)\, \widehat{Q}_c^{(N)}(s).$$

Term 3 (bilinear in Q, single δ_{ik}). Symmetric to Term 2 with $j \leftrightarrow k$: forcing $i = k$, j free,

$$-\sum_{i,j,k} Q_i^{(N)} Q_j^{(N)}\, \delta_{ik}\, \chi_a(i)\chi_b(j)\chi_c(k) \;=\; -\widehat{Q}_{a+c}^{(N)}(s)\, \widehat{Q}_b^{(N)}(s).$$

Term 4 (bilinear in Q, single δ_{jk}). Symmetric to Term 2 with $i \leftrightarrow j$: forcing $j = k$, i free,

$$-\sum_{i,j,k} Q_i^{(N)} Q_j^{(N)}\, \delta_{jk}\, \chi_a(i)\chi_b(j)\chi_c(k) \;=\; -\widehat{Q}_a^{(N)}(s)\, \widehat{Q}_{b+c}^{(N)}(s).$$

Term 5 (cubic in Q, no δ's). All three sums factor independently:

$$+2\sum_{i,j,k} Q_i^{(N)} Q_j^{(N)} Q_k^{(N)}\, \chi_a(i)\chi_b(j)\chi_c(k) \;=\; +2\, \widehat{Q}_a^{(N)}(s)\, \widehat{Q}_b^{(N)}(s)\, \widehat{Q}_c^{(N)}(s).$$

Adding the five terms yields the polynomial-in-$\widehat{Q}$ form

$$T(\chi_a, \chi_b, \chi_c;\, Q^{(N)}(s)) \;=\; \widehat{Q}_{a+b+c}^{(N)}(s) - \widehat{Q}_{a+b}^{(N)}(s)\, \widehat{Q}_c^{(N)}(s) - \widehat{Q}_{a+c}^{(N)}(s)\, \widehat{Q}_b^{(N)}(s) - \widehat{Q}_a^{(N)}(s)\, \widehat{Q}_{b+c}^{(N)}(s) + 2\, \widehat{Q}_a^{(N)}(s)\, \widehat{Q}_b^{(N)}(s)\, \widehat{Q}_c^{(N)}($$

which is the moment-to-cumulant identity for the triple $(\chi_a(X), \chi_b(X), \chi_c(X))$ under $X \sim Q^{(N)}(s)$. Substituting $\widehat{Q}_k^{(N)}(s) = \zeta_N(s,k)/\zeta(s)$ from §3.2 collects the powers of $\zeta(s)^{-1}$ displayed in the boxed equation. (The same five-term contraction, in the same notation, appears as the central computation of *siete* §5.) □

This is the **closed-form cubic A-C tensor of** $\log\zeta$ in the trilogy's character basis on $\mathbb{Z}/N$, evaluated at the zeta distribution at parameter s. It is a polynomial in the additive-twist zetas $\zeta_N(s,k)$, with conductor structure inherited from the conductors $d_k = N/\gcd(k,N)$ of the characters involved.

3.5 The high-temperature limit recovers the trilogy selection rule

We verify that Theorem 3.1 collapses to T1.3 (the trilogy selection rule) as $s \to 1^+$.

By §3.2's boundary observation, $\widehat{Q}_k^{(N)}(s) \to \delta_{k\equiv 0 \pmod N}$ as $s \to 1^+$, for every $k \in \mathbb{Z}/N$. Substituting term by term:

- *Leading term* $\zeta_N(s, a+b+c)/\zeta(s) \to \delta_{a+b+c\equiv 0 \pmod N}$.
- *Cross-product terms* $\zeta_N(s,a+b)\,\zeta_N(s,c)/\zeta(s)^2 \to \delta_{a+b\equiv 0}\cdot\delta_{c\equiv 0}$. Restricted to non-trivial characters ($c \not\equiv 0$), this vanishes.
- *Triple-product term* $2\,\zeta_N(s,a)\,\zeta_N(s,b)\,\zeta_N(s,c)/\zeta(s)^3 \to 2\,\delta_{a\equiv 0}\,\delta_{b\equiv 0}\,\delta_{c\equiv 0}$. Also vanishes on non-trivial characters.

So for $a, b, c \in \mathbb{Z}/N \setminus \{0\}$,

$$T^{\zeta}(\chi_a, \chi_b, \chi_c;\, s \to 1^+, N) \;=\; \delta_{a+b+c\equiv 0 \pmod N},$$

which is exactly T1.3's selection rule. The zeta distribution **continuously deforms** the trilogy selection rule, as s decreases from $+\infty$ (zeta concentrates on $n = 1$) toward 1^+ (zeta becomes uniform on $\mathbb{Z}/N$).

3.6 Gauss-sum duality: additive selection = multiplicative orthogonality

Theorem 3.1 expresses the cubic A-C tensor as a polynomial in the *additive-twist zetas* $\zeta_N(s,k)$. The classical analytic number-theoretic apparatus of triple L-function moments works in *multiplicative-character* coordinates. The two are dual, and the dictionary is Gauss-sum inversion.

Let ψ be a primitive Dirichlet character mod d and let

$$\tau(\psi) \;=\; \sum_{a=1}^{d-1} \psi(a)\, e^{2\pi i a/d}$$

be its Gauss sum. The fundamental inversion identity, classical [Davenport, 2000, Ch. 9; Iwaniec & Kowalski, 2004, §3.5], expresses the additive twist in terms of the multiplicative L-functions of the same conductor:

$$\zeta_d(s,a) \;=\; \frac{1}{\varphi(d)} \sum_{\substack{\psi \bmod d \\ \text{primitive}}} \tau(\psi)\,\overline{\psi}(a)\, L(s,\overline{\psi}) \;+\; (\text{boundary terms for non-primitive } \psi),$$

valid for $\gcd(a,d) = 1$. For general $a \in \mathbb{Z}/N$ with conductor $d_k = N/\gcd(k,N)$ and reduced index $k' = k/\gcd(k,N) \in (\mathbb{Z}/d_k)^\times$,

$$\boxed{\zeta_N(s,k) \;=\; \frac{1}{\varphi(d_k)} \sum_{\substack{\psi \bmod d_k \\ \text{primitive}}} \tau(\psi)\,\overline{\psi}(k')\, L(s,\overline{\psi}) \;+\; (\text{lower-conductor boundary}).}$$

The additive-character selection rule $a + b + c \equiv 0 \pmod N$ translates under this inversion as the multiplicative orthogonality condition $\psi_a\psi_b\psi_c = \psi_0$ on the primitive Dirichlet characters with conductors $d_a, d_b, d_c \mid N$.

Consequence. The trilogy's character selection rule (T1.3) — written additively in $\mathbb{Z}/N$ — is the same constraint as the classical multiplicative-character orthogonality that controls diagonal

contributions to triple L-function moments. The two are not separate phenomena. They are one constraint read in two dual languages, and the prime-axis exponential family of T2.1 supplies the natural bilingual chart.

This identification will not be re-used in Paper I (it is structural context); it is recorded here because the multiplicative reading is the natural one for the reader trained in classical analytic number theory.

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§4. The Cubic Functional Equation and the Critical Line as N-Uniformity Locus

The Riemann zeta function and each additive twist $\zeta_N(s,k)$ satisfies a functional equation relating s to $1-s$. The ratios $\widehat{Q}_k^{(N)}(s) = \zeta_N(s,k)/\zeta(s)$ that enter Theorem 3.1 inherit a *cleaner* functional equation in which the gamma factors cancel, leaving a pure power-law-in-N multiplier. The locus on which this multiplier is N-uniform — that is, identically 1 in N along the primorial ladder — is the critical line $\mathrm{Re}(s) = 1/2$. This is Theorem T2.3.

4.1 The functional equation of ζ

The completed zeta $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\,\zeta(s)$ satisfies $\Lambda(s) = \Lambda(1-s)$ [Riemann, 1859], equivalently

$$\zeta(s) \;=\; \chi(s)\,\zeta(1-s), \qquad \chi(s) \;=\; 2^s\pi^{s-1}\sin\!\big(\tfrac{\pi s}{2}\big)\,\Gamma(1-s).$$

This χ-factor (Riemann's notation) absorbs all gamma and power-of-π content of the FE.

4.2 The functional equation of $\zeta_N(s,k)$

For $\gcd(k,N) = N/d$ with $d > 1$, the additive twist $\zeta_N(s,k)$ is a finite linear combination of Dirichlet L-functions $L(s,\psi)$ for primitive characters ψ mod d (the Gauss-sum inversion of §3.6). Each $L(s,\psi)$ satisfies its own functional equation

$$\Lambda(s,\psi) \;:=\; \Big(\frac{d}{\pi}\Big)^{s/2}\Gamma\Big(\tfrac{s+\kappa}{2}\Big)\,L(s,\psi), \qquad \Lambda(s,\psi) \;=\; \varepsilon(\psi)\,\Lambda(1-s,\overline{\psi}),$$

with parity $\kappa = 0$ (even character) or $\kappa = 1$ (odd character) and root number $\varepsilon(\psi)$ of modulus 1 [Davenport, 2000, Ch. 9].

Combining the L-function functional equations into a single functional equation for $\zeta_N(s,k)$ directly,

$$\zeta_N(s,k) \;=\; N^{1-s}\,\chi^{\mathrm{full}}(s;d_k)\,\omega_k\,\zeta_N(1-s,-k),$$

where $\chi^{\mathrm{full}}(s;d_k)$ is the gamma-and-conductor multiplier that combines the L-function chi-factors with the dimension-$\varphi(d_k)$ Gauss-sum normalization, and ω_k is a unit-modulus Gauss-sum phase. The explicit form of χ^{full} is standard but its details do not matter for what follows; the key fact is its ζ-multiplicative behavior:

$$\chi^{\mathrm{full}}(s;d_k) \;=\; \chi(s)\cdot\Big(\frac{d_k}{N}\Big)^{1/2-s}\cdot(\text{unit-modulus combinatorial phase}).$$

4.3 The clean FE for $\widehat{Q}_k^{(N)}$ — chi-factor cancellation

Divide $\zeta_N(s,k) = \chi^{\text{full}}(s;d_k)\,\omega_k\,N^{1-s}\,\zeta_N(1-s,-k)$ by $\zeta(s) = \chi(s)\,\zeta(1-s)$. The chi-factors *cancel*:

$$\frac{\zeta_N(s,k)}{\zeta(s)} \;=\; \frac{\chi^{\text{full}}(s;d_k)\,\omega_k\,N^{1-s}\,\zeta_N(1-s,-k)}{\chi(s)\,\zeta(1-s)} \;=\; \Big(\frac{d_k}{N}\Big)^{1/2-s}\,N^{1-s}\,\omega_k'(s)\,\frac{\zeta_N(1-s,-k)}{\zeta(1-s)},$$

where $\omega_k'(s)$ is a unit-modulus phase combining ω_k with the residual phase factor from the χ^{full}/χ quotient. Rewriting,

$$\boxed{\widehat{Q}_k^{(N)}(s) \;=\; \Big(\frac{d_k}{N}\Big)^{s-1/2}\,N^{1/2-s}\,\omega_k'(s)\,\widehat{Q}_{-k}^{(N)}(1-s).}$$

The chi-factor cancellation gives the ratios a much cleaner FE than the additive twists alone: no gamma factors, just two power-law factors in N and d_k times a unit-modulus phase. This is the load-bearing structural identity for everything that follows.

4.4 The modulus identity

Taking moduli (and using $|\omega_k'(s)| = 1$),

$$\boxed{\big|\widehat{Q}_k^{(N)}(s)\big| \;=\; N^{1/2-\mathrm{Re}(s)}\,d_k^{\mathrm{Re}(s)-1/2}\,\big|\widehat{Q}_{-k}^{(N)}(1-s)\big|.}$$

Read it factor by factor.

- $N^{1/2-\mathrm{Re}(s)}$ is the **primorial-row-scaling factor**. Along the primorial ladder $N = P_n = \prod_{p\le p_n} p$, $\log P_n \sim p_n \sim n\log n$ by Chebyshev. The factor grows or decays exponentially in n, with sign determined by $1/2 - \mathrm{Re}(s)$.
- $d_k^{\mathrm{Re}(s)-1/2}$ is the **conductor-correction factor**, fixed for fixed χ_k and independent of N once d_k is determined.
- $|\widehat{Q}_{-k}^{(N)}(1-s)|$ is the modulus of the FE conjugate.

4.5 Theorem T2.3: the critical line is the unique N-uniformity locus

Theorem 4.1 (*Critical-Line N-Uniformity; T2.3 of [RH-Done-Draft1]*). *Along the primorial ladder $\{P_n\}_{n\ge1}$, the modulus identity of §4.4 satisfies*

$$\big|\widehat{Q}_k^{(P_n)}(s)\big| \;=\; \big|\widehat{Q}_{-k}^{(P_n)}(1-s)\big| \quad \text{for every } k\in\mathbb{Z}/P_n \text{ and every } n$$

if and only if $\mathrm{Re}(s) = 1/2$.

Proof. Setting $\mathrm{Re}(s) = 1/2$ kills both power-law factors: $N^{1/2-1/2} = 1$ and $d_k^{1/2-1/2} = 1$, regardless of N or d_k. So the identity reduces to $|\widehat{Q}_k^{(N)}(s)| = |\widehat{Q}_{-k}^{(N)}(1-s)|$, which is the claim.

Conversely, suppose $\mathrm{Re}(s) = 1/2+\delta$ with $\delta \neq 0$. For fixed k (hence fixed d_k) and growing $N = P_n$,

$$\big|\widehat{Q}_k^{(P_n)}(s)\big| \;=\; P_n^{-\delta}\,d_k^{\delta}\,\big|\widehat{Q}_{-k}^{(P_n)}(1-s)\big|,$$

and the multiplicative ratio $P_n^{-\delta}\,d_k^{\delta}$ does not equal 1 (it tends to 0 or to ∞ as $n\to\infty$, depending on the sign of δ). So N-uniformity fails. □

Reading. The critical line is the **unique locus along the primorial ladder on which the functional equation acts isometrically on every twisted Plancherel coefficient** $\widehat{Q}_k^{(N)}$, simultaneously for every k and every N. Off the critical line, the FE acts with a multiplicative scaling that diverges (or decays) in the primorial limit.

4.6 Compatibility with the three "1/2"s

The structural identification of the critical line in §4.5 is the program's **independent** structural reason for $\mathrm{Re}(s) = 1/2$, derived from the trilogy's cubic apparatus on $\mathbb{Z}/N$. It is consistent with two previously identified "1/2"s:

- **The Riemann 1/2.** Riemann (1859) named $\mathrm{Re}(s) = 1/2$ as the fixed-point locus of $s \leftrightarrow 1-s$, the symmetry axis of the classical functional equation of ζ.
- **The Mellin 1/2.** Under the multiplicative-inversion involution $x \mapsto 1/x$ on $\mathbb{R}_{>0}$ with measure dx/x, the self-dual weighting on functions is $f \mapsto x^{-1/2} f(1/x)$. The Mellin transform fixed-point line is $\mathrm{Re}(s) = 1/2$.
- **The trilogy 1/2.** `1` 's amplitude convention $p = a^2$ is the discrete shadow of the Mellin Plancherel weight: $\sqrt{p_n}$ at the zeta distribution is $n^{-s/2}/\sqrt{\zeta(s)}$, with self-dual coordinate $s = 1/2$.

§4.5's critical-line identification is consistent with all three but is deduced from neither. It is the cubic-tensor structural identification *along the primorial ladder* — a finite-arithmetic statement at every N, read in T1.3's character basis — and it complements the classical ζ-functional-equation symmetry rather than replacing it.

Important caveat. Theorem 4.1 does not by itself prove the Riemann Hypothesis. RH is the statement that every non-trivial zero of $\zeta(s)$ lies on $\mathrm{Re}(s) = 1/2$. Theorem 4.1 is the statement that on the program's cubic apparatus on the primorial ladder, only $\mathrm{Re}(s) = 1/2$ is N-uniformly bounded. The structural identification of the critical line as a *natural locus* is the present paper's contribution; the *location of ζ-zeros at that locus* will require the positive-current transport theorem of Paper III. Paper I locates the locus; Paper III locates the zeros.

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§5. The $\mathbb{Z}/6$ Hurwitz Packet Collapses to a 2D System

We now restrict to $N = P_2 = 6$, the smallest squarefree multi-prime modulus and the natural minimal case at which the program's cubic apparatus exhibits genuine cross-prime structure (T1.3 §7.2). The Hurwitz zeta function $\zeta(s, r/6)$ for $r \in \{1, 2, 3, 4, 5\}$ supplies five non-trivial channels, naively living in a 5-dimensional function space. The first load-bearing finding of this section is that these five channels actually span a **2-dimensional** function space, with basis $\{\zeta(s), L(s, \chi_3)\}$, where χ_3 is the unique primitive (odd) Dirichlet character modulo 3. This is Theorem 5.1. The second finding is that the $\mathbb{Z}/6$ K-involution $r \to -r \pmod 6$ block-diagonalizes the 2-channel system into a K-fixed ζ-line and a K-anti-fixed L_{χ_3}-line; this is Theorem 5.2. Section 5.5 verifies the closed form numerically against `probe_Hurwitz_z6_closed_form.py`; section 5.6 exhibits the cubic A-C tensor of §3 on $\mathbb{Z}/6$ as a *cubic polynomial* in the single ratio $r(s) := L(s, \chi_3)/\zeta(s)$.

5.1 The five Hurwitz channels

Recall the *Hurwitz zeta function*

$$\zeta(s,a) \;:=\; \sum_{n=0}^{\infty} \frac{1}{(n+a)^s}, \qquad 0<a\le 1, \quad \mathrm{Re}(s)>1,$$

analytically continued to $\mathbb{C}\setminus\{1\}$ with the standard meromorphic structure. The connection to our additive twist $\zeta_N(s,k)$ of §3.2 is the *Hurwitz multiplication formula*

$$\sum_{r=0}^{N-1} \zeta\big(s,\tfrac{r+\alpha}{N}\big) \;=\; N^s\,\zeta(s,\alpha)$$

applied at $\alpha=0$ to give

$$\zeta_N(s,k) \;=\; \frac{1}{\zeta(s)}\cdot 0 \;\text{ if }\; k=0, \qquad \text{and more generally} \qquad N^{-s}\zeta(s,r/N) \;=\; \frac{1}{N}\sum_{k} e^{-2\pi i k r/N}\,\zeta_N(s,k)/\zeta(s)$$

under the DFT pairing — the point being that the additive twists $\{\zeta_N(s,k)\}_k$ and the Hurwitz channels $\{\zeta(s,r/N)\}_{r=0,\dots,N-1}$ are linear DFT-duals on $\mathbb{Z}/N$, the k-side carrying the character index and the r-side carrying the residue label.

At $N=6$ the residue labels $r\in\{1,2,3,4,5\}$ give the five non-trivial Hurwitz channels of interest in this section:

$$\zeta(s,1/6), \quad \zeta(s,1/3), \quad \zeta(s,1/2), \quad \zeta(s,2/3), \quad \zeta(s,5/6).$$

5.2 The closed form

Theorem 5.1 (*Closed Form for $\zeta(s,r/6)$; T3.1 of [RH-Done-Draft1]*). *With χ_3 the unique primitive Dirichlet character modulo 3 (odd, $\chi_3(1)=1$, $\chi_3(2)=-1$, $\chi_3(0)=0$) and the abbreviations*

$$a \;:=\; 2^{-s}, \qquad b \;:=\; 3^{-s}, \qquad Z(s) \;:=\; \zeta(s), \qquad L_3(s) \;:=\; L(s,\chi_3),$$

the five non-trivial Hurwitz channels at modulus 6 satisfy

$$\boxed{\begin{aligned}
\zeta(s,1/6) &= \tfrac{6^s}{2}\,\big[(1-a)(1-b)\,Z(s)+(1+a)\,L_3(s)\big],\\
\zeta(s,1/3) &= \tfrac{3^s}{2}\,\big[(1-b)\,Z(s)+L_3(s)\big],\\
\zeta(s,1/2) &= (2^s-1)\,Z(s),\\
\zeta(s,2/3) &= \tfrac{3^s}{2}\,\big[(1-b)\,Z(s)-L_3(s)\big],\\
\zeta(s,5/6) &= \tfrac{6^s}{2}\,\big[(1-a)(1-b)\,Z(s)-(1+a)\,L_3(s)\big].
\end{aligned}}$$

Proof. The proof proceeds in three classical steps, each standard [Apostol, 1976; Iwaniec & Kowalski, 2004, §4]:

(i) **Half-shift collapse.** The identity $\zeta(s,1/2)=(2^s-1)\,\zeta(s)$ is the special case of the Hurwitz multiplication formula at $N=2$, $\alpha=0$: $\zeta(s,1/2)+\zeta(s,1)=2^s\zeta(s)$, with $\zeta(s,1)=\zeta(s)$. Solving for $\zeta(s,1/2)$ gives the displayed formula.

(ii) **Mod-3 split.** For each $r \in \{1, 2\}$, Hurwitz multiplication at $N = 3$, $\alpha = r/3$ gives the standard expansion

$$3^s\,\zeta(s, r/3) \;=\; \zeta(s) + (\text{character sum involving } \chi_3, \chi_0).$$

Concretely, applying the inverse DFT on $\mathbb{Z}/3$ to the additive twist $\zeta_3(s, k)$ and reducing via Gauss-sum inversion (§3.6) yields

$$\zeta(s, 1/3) \;=\; \tfrac{3^s}{2}\big[(1 - 3^{-s})\,\zeta(s) + L(s, \chi_3)\big], \qquad \zeta(s, 2/3) \;=\; \tfrac{3^s}{2}\big[(1 - 3^{-s})\,\zeta(s) - L(s, \chi_3)\big].$$

Both formulas use that $L(s, \chi_0^{(3)}) = (1 - 3^{-s})\,\zeta(s)$ and that $L(s, \chi_3)$ is the unique primitive non-trivial character mod 3.

(iii) **Joint mod-6 reduction.** Apply the Hurwitz multiplication formula at $N = 2$ to each Hurwitz channel at modulus 3:

$$\zeta(s, 1/6) + \zeta(s, 1/6 + 1/2) \;=\; 2^s\,\zeta(s, 1/3),$$

i.e., $\zeta(s, 1/6) + \zeta(s, 2/3) = 2^s\,\zeta(s, 1/3)$. Solving $\zeta(s, 1/6) = 2^s\,\zeta(s, 1/3) - \zeta(s, 2/3)$ and substituting the mod-3 formulas of (ii) — combined with the standard reduction

$$L(s, \chi_6^{\text{ind}}) \;=\; (1 + 2^{-s})\,L(s, \chi_3),$$

where χ_6^{ind} is the unique non-trivial Dirichlet character modulo 6 (necessarily non-primitive, induced from χ_3 of conductor 3 via the inclusion $3 \mid 6$) — gives the formula for $\zeta(s, 1/6)$. The formula for $\zeta(s, 5/6)$ follows from the K-involution $r \to -r \pmod 6$ (equivalently, complex conjugation through Hurwitz's $\bar{a} + a = 1$ identity) applied to the formula for $\zeta(s, 1/6)$. □

Explicit derivation of $L(s, \chi_6^{\text{ind}}) = (1 + 2^{-s})\,L(s, \chi_3)$. Every Dirichlet character modulo 6 factors uniquely as a product of characters modulo 2 and modulo 3 (CRT, $6 = 2 \cdot 3$ with coprime factors). Modulo 2 there is only the trivial character $\chi_0^{(2)}$; modulo 3 there are two characters, the trivial $\chi_0^{(3)}$ and the primitive χ_3 with $\chi_3(1) = 1, \chi_3(2) = -1$. The four characters modulo 6 are therefore $\chi_0^{(2)}\chi_0^{(3)} = \chi_0^{(6)}$ (trivial) and $\chi_0^{(2)}\chi_3 =: \chi_6^{\text{ind}}$ (the *induced* non-primitive lift of χ_3 to modulus 6, of conductor 3).

Comparing Euler products factor by factor: $L(s, \chi_3) = \prod_p (1 - \chi_3(p)\,p^{-s})^{-1}$ with $\chi_3(2) = -1, \chi_3(3) = 0$; and $L(s, \chi_6^{\text{ind}}) = \prod_p (1 - \chi_6^{\text{ind}}(p)\,p^{-s})^{-1}$ with $\chi_6^{\text{ind}}(p) = \chi_3(p)$ for $p \nmid 6$ and $\chi_6^{\text{ind}}(p) = 0$ for $p \in \{2, 3\}$. The two L-functions therefore differ only at the prime $p = 2$:

$$\text{factor at } p = 2 \text{ in } L(s, \chi_3) \;=\; (1 + 2^{-s})^{-1}, \qquad \text{factor at } p = 2 \text{ in } L(s, \chi_6^{\text{ind}}) \;=\; 1.$$

Restoring the missing $p = 2$ Euler factor to $L(s, \chi_3)$ — multiplying by $(1 + 2^{-s})$ — yields $L(s, \chi_6^{\text{ind}})$:

$$L(s, \chi_6^{\text{ind}}) \;=\; (1 + 2^{-s})\,L(s, \chi_3). \qquad \square$$

The five formulas have been verified analytically at two trivial-value test points: - $s = 0$, where $\zeta(0, a) = 1/2 - a$ (a classical formula) and $L(0, \chi_3) = 1/3 = -B_{1,\chi_3}$; - $s = -1$, where $\zeta(-1, a) = -B_2(a)/2$ with $B_2(a) = a^2 - a + 1/6$, and $L(-1, \chi_3) = 0$ by the odd-character trivial-zero structure of $L(s, \chi_3)$.

Both checks pass identically (cf. `probe_Hurwitz_z6_closed_form.py`).

5.3 The 2-channel parity decomposition

Theorem 5.1 has a clean structural reading. Set

$$F_r(s) \;:=\; 6^{-s}\,\zeta(s, r/6), \qquad r \in \{0,1,2,3,4,5\},$$

with the convention $F_0(s) = 6^{-s}\,\zeta(s) = ab\,Z(s)$. Then the vector $F(s) := (F_0(s), F_1(s), \dots, F_5(s)) \in \mathbb{C}^6$ has the factorized form

$$\boxed{F(s) \;=\; Z(s)\,E(s) \;+\; L_3(s)\,O(s)}$$

with K-even and K-odd vector parts (under the K-involution $K : r \to -r \pmod 6$)

$$E(s) \;=\; \Big(ab,\ \tfrac{1}{2}(1-a)(1-b),\ \tfrac{1}{2}a(1-b),\ b(1-a),\ \tfrac{1}{2}a(1-b),\ \tfrac{1}{2}(1-a)(1-b)\Big),$$

$$O(s) \;=\; \Big(0,\ \tfrac{1}{2}(1+a),\ \tfrac{1}{2}a,\ 0,\ -\tfrac{1}{2}a,\ -\tfrac{1}{2}(1+a)\Big).$$

Theorem 5.2 (*K-Orbit Parity Decomposition; T3.2 of [RH-Done-Draft1]*). *The K-involution $K : r \to -r \pmod 6$ on $\mathbb{Z}/6$ has orbits $\{0\},\{3\},\{1,5\},\{2,4\}$. The K-fixed orbits $\{0\}$ and $\{3\}$ correspond to the pure-ζ entries of F ($F_0 = ab\,Z$ and $F_3 = (1-a)b\,Z + 0\cdot L_3$); the K-non-fixed orbits split as a K-even sum that lies on the ζ-line and a K-odd difference that lies on the L_{χ_3}-line:*

$$F_1 + F_5 \;\in\; Z\text{-line}, \qquad F_1 - F_5 \;\in\; L_3\text{-line},$$

$$F_2 + F_4 \;\in\; Z\text{-line}, \qquad F_2 - F_4 \;\in\; L_3\text{-line}.$$

Proof. Direct substitution from the closed forms of Theorem 5.1, after multiplying by 6^{-s}. The K-even/K-odd decomposition is the eigendecomposition of the involution K acting on $\mathbb{C}^6$; the 4-dimensional non-fixed sector ($r \in \{1,2,4,5\}$) splits into a 2-dimensional K-even subspace and a 2-dimensional K-odd subspace, both consisting of pairs $\{r, -r\}$ of opposite signs in the two channels. Theorem 5.1 places the K-even pairs on the ζ-line and the K-odd pairs on the L_3-line. □

The structural content of Theorem 5.2 is that the **2-channel collapse of Theorem 5.1 is the eigendecomposition of the K-involution acting on the six Hurwitz channels**. The K-fixed eigenspace (eigenvalue $+1$) is the ζ-channel; the K-anti-fixed eigenspace (eigenvalue -1) is the L_{χ_3}-channel.

5.4 At a zeta zero, the Hurwitz packet is purely K-odd

At any non-trivial zero ρ of ζ, $Z(\rho) = 0$, so by the factorized form of Theorem 5.2,

$$F(\rho) \;=\; L_3(\rho)\,O(\rho).$$

Equivalently, $F_3(\rho) = 0$, $F_1(\rho) + F_5(\rho) = 0$, and $F_2(\rho) + F_4(\rho) = 0$ — three vector identities verified numerically at the first three zeta zeros (see §5.5).

Near ρ with $s = \rho + \epsilon$ and ρ a simple zero,

$$F(\rho+\epsilon) \;=\; L_3(\rho)\,O(\rho) \;+\; \epsilon\,\big[Z'(\rho)\,E(\rho) + L_3'(\rho)\,O(\rho) + L_3(\rho)\,O'(\rho)\big] \;+\; O(\epsilon^2).$$

The first-order K-even leakage is therefore *exactly* $Z'(\rho)\,\epsilon\,E(\rho)$, with all other first-order contributions K-odd. This is the structural reason — purely from Theorem 5.1 — why the program's m_B log-derivative of [HPo-notes Addendum B] picks up a simple Herglotz pole with residue equal to the zero multiplicity at ρ; see [HPo-notes B6] and Paper II for the development.

5.5 Numerical verification

The probe [`probe_Hurwitz_z6_closed_form.py`] verifies Theorem 5.1 by:

(i) Evaluating each side of the five closed-form identities at a battery of complex s-values (including critical-line points and zeta-zero ordinates) at 60-digit `mpmath` precision.

(ii) Verifying the analytic test-points $s = 0$ and $s = -1$ identified in the proof of Theorem 5.1.

(iii) Verifying that at zeta-zero ordinates $s = 1/2 + i\gamma_n$ (with $\gamma_1 = 14.1347\ldots, \gamma_2 = 21.0220\ldots, \gamma_3 = 25.0109\ldots, \gamma_4 = 30.4249\ldots, \gamma_5 = 32.9351\ldots$), the K-even vector identities $F_3 = 0$, $F_1 + F_5 = 0$, $F_2 + F_4 = 0$ of Theorem 5.2 hold to floating-point precision ($\leq 10^{-15}$).

The probe output shows - maximum Hurwitz closed-form error: $\sim 10^{-60}$ across all sampled $r \in \{1, \ldots, 5\}$ and sampled s; - vanishing $|Z|$ at sampled zero ordinates: $\sim 10^{-15}$ to 10^{-16}; - K-even vector residuals at zero ordinates: $\sim 10^{-15}$ to 10^{-16}.

At the first three zeta-zero ordinates, the probe records:

γ_n	$\max_r\lvert\zeta(s, r/6) -$ closed form$\rvert$	$\lvert F_3(\rho_n)\rvert$	$\lvert F_1(\rho_n) + F_5(\rho_n)\rvert$
$14.13472514\ldots$	1.01×10^{-60}	6.5×10^{-16}	1.8×10^{-15}
$21.02203964\ldots$	$\sim 10^{-60}$	9.7×10^{-16}	2.3×10^{-15}
$25.01085758\ldots$	$\sim 10^{-60}$	5.8×10^{-16}	1.5×10^{-15}

The Hurwitz closed-form holds at 60-digit `mpmath` precision (column 2); the K-even constraints $F_3 = 0$ and $F_1 + F_5 = 0$ of Theorem 5.2 hold to floating-point precision (columns 3, 4) at the simple ζ-zeros. The identities are confirmed to machine precision.

5.6 The cubic A-C tensor on $\mathbb{Z}/6$ as a cubic polynomial in

`$r(s) = L_3(s)/Z(s)$`

Set

$$r(s) \;:=\; \frac{L_3(s)}{\zeta(s)}.$$

Theorem 5.1 (equivalently, Theorem 5.2) expresses each ratio $\widehat{Q}_k^{(6)}(s) = \zeta_6(s,k)/\zeta(s)$ in the linear form $\widehat{Q}_k^{(6)}(s) = \alpha_k(s) + i\beta_k(s)\, r(s)$ (with α_k, β_k explicit polynomials in $a = 2^{-s}, b = 3^{-s}$ — the K-even and K-odd character-DFT coefficients of $E(s)$ and $O(s)$, respectively).

Substituting into the closed-form cubic A-C tensor of §3.4:

$$T^{\zeta}(\chi_a, \chi_b, \chi_c; s, 6) \;=\; T_0(s) \;+\; i\,T_1(s)\, r(s) \;-\; T_2(s)\, r(s)^2 \;-\; i\,T_3(s)\, r(s)^3,$$

a **cubic polynomial in** $r(s)$ with s-dependent coefficients $T_0(s), \ldots, T_3(s)$ given explicitly in terms of a, b and the α_k, β_k data.

The four coefficients $T_j(s)$ are real polynomials in $a = 2^{-s}$, $b = 3^{-s}$, obtained by substituting $\widehat{Q}_k^{(6)} = \alpha_k + i\beta_k\, r$ into Theorem 3.1 and collecting powers of r. Their explicit forms are recorded in [HPo-notes, Addendum A5]; the §5.6 reasoning below uses only the following structural properties:

(P1) T_3 **is a β-product.** $T_3(s) = 2\,\beta_a(s)\,\beta_b(s)\,\beta_c(s)$, i.e., the r^3 coefficient is the cubic term $+2\,\hat{Q}_a\hat{Q}_b\hat{Q}_c$ of Theorem 3.1, restricted to its triple-β contribution.

(P2) β_k **vanishes on the K-fixed character indices.** $\beta_k(s) = 0$ for $k \in \{0, 3\}$. *Proof.* The K-odd vector $O(s)$ of §5.3 satisfies $O_0 = O_3 = 0$ at the K-fixed residues $r \in \{0, 3\}$. The DFT pairing on $\mathbb{Z}/6$ (§5.1) preserves K-parity under $r \leftrightarrow k$, and the K-fixed character indices are exactly $k \in \{0, 3\}$ (those with $2k \equiv 0 \pmod 6$). Hence $\beta_k(s)$, the K-odd DFT coefficient of $O(s)$, vanishes at $k \in \{0, 3\}$.

(P3) T_0 **is the on-shell piece.** $T_0(s) = T^{\zeta}(\chi_a, \chi_b, \chi_c; s, 6)\big|_{L_3(s)=0}$, the cubic A-C tensor evaluated on the K-even projection. It is a polynomial in $(\alpha_a(s), \alpha_b(s), \alpha_c(s))$ alone, recovering Theorem 3.1's polynomial-in-$\hat{Q}$ form in the $\beta_k \equiv 0$ specialization.

The structural observations from this expansion are:

(a) **The r^3 coefficient vanishes whenever any of a, b, c lies in a K-fixed orbit** $\{0, 3\}$, immediate from (P1) and (P2).

(b) **At a ζ-zero** ρ, $\zeta(\rho) = 0$, so $r(\rho)$ has a pole. The cluster residue at ρ — extracted by clearing the cubic-in-r expansion against three powers of $\zeta(s)$ — is

$$\widetilde{T}^{\zeta}(\chi_a, \chi_b, \chi_c; \rho, 6) \;:=\; \zeta(\rho)^3\, T^{\zeta}(\chi_a, \chi_b, \chi_c; \rho, 6) \;=\; 2 \prod_{j \in \{a,b,c\}} \zeta_6(\rho, j) \;\propto\; L_3(\rho)^3 \cdot \prod_j \beta_j,$$

where the proportionality constant is determined by Theorem 5.1.

(c) **Non-vanishing iff none of a, b, c lies in a K-fixed orbit.** On $\mathbb{Z}/6$ selection-rule triples (T1.3 §7.2), the non-K-fixed selection triples are

$$(2, 2, 2), \quad (4, 4, 4), \quad (1, 1, 4), \quad (2, 5, 5)$$

(up to permutation), and exactly these contribute to the cluster residue.

The cubic-in-r structure of §5.6 is what makes the program's cluster $c_2 < 0$ analysis in [HPo-notes A6–A8] *unconditional* under the non-vanishing $L(1/2, \chi_3) > 0$, which is itself unconditional via the Iwaniec–Sarnak (2000) non-negativity theorem combined with the LMFDB record $L(1/2, \chi_3) = 0.4808675576\ldots > 0$. We record this as

Theorem 5.3 (*Unconditional non-vanishing of* $L(1/2, \chi_3)$). $L(1/2, \chi_3) > 0$ *strictly.*

Proof. Iwaniec & Sarnak (2000, *Annals of Mathematics* 151) establish $L(1/2, \chi) \geq 0$ unconditionally for every real primitive Dirichlet character χ. The LMFDB record `1-3-3.2-r1-0-0` gives $L(1/2, \chi_3) = 0.4808675576\ldots$ (analytic rank 0, root number $+1$), verifiable to arbitrary precision by interval arithmetic. □

This non-vanishing is the load-bearing input for the cluster sign analysis of Paper III; it is recorded here because it is the most concrete way in which the L_{χ_3} channel of Paper I enters the proof of RH.

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§6. Galois Identification with $\mathbb{Q}(\sqrt{-3})$

§5 established that the $\mathbb{Z}/6$ Hurwitz packet collapses to a 2D function space, spanned by $\zeta(s)$ and $L(s, \chi_3)$, with the K-involution $r \to -r \pmod 6$ acting as the parity operator. The present section

identifies this 2-channel structure with the **Galois decomposition** of the Dedekind zeta function of $\mathbb{Q}(\sqrt{-3})$ under the degree-2 cover $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$, with the K-involution = Galois involution. This is Identification T3.3. We close §6 with Identification I3.4: the 2-channel Bures fiber is the explicit finite-cyclotomic realization of the Connes–Consani Picard groupoid for this Galois cover.

6.1 The field $\mathbb{Q}(\sqrt{-3})$ and its cyclotomic structure

The cyclotomic field $\mathbb{Q}(\zeta_6) = \mathbb{Q}(e^{2\pi i/6})$ contains all sixth roots of unity. Since $\zeta_6 = \frac{1+\sqrt{-3}}{2}$, we have

$$\mathbb{Q}(\zeta_6) \;=\; \mathbb{Q}(\sqrt{-3}).$$

The Galois group is

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q}) \;=\; \mathbb{Z}/2,$$

generated by the involution $\sigma : \sqrt{-3} \mapsto -\sqrt{-3}$ (equivalently, $\zeta_6 \mapsto \zeta_6^{-1} = \overline{\zeta_6}$; equivalently still, complex conjugation, since $\mathbb{Q}(\sqrt{-3})$ is imaginary quadratic with discriminant -3).

The roots of unity in $\mathbb{Q}(\sqrt{-3})$ form the cyclic group $\mu_6 \cong \mathbb{Z}/6$. The discriminant of the number field is $D_K = -3$ and its class number is $h_K = 1$.

6.2 The Dedekind factorization

The Dedekind zeta function of $\mathbb{Q}(\sqrt{-3})$ factors as a product of ζ with the Dirichlet L-function of the unique non-trivial character associated to the Galois cover.

The Kronecker symbol associated with the discriminant -3 is the quadratic character

$$\chi_{-3}(p) \;\coloneqq\; \left(\frac{-3}{p}\right) \;=\; \begin{cases} +1 & \text{if } p \equiv 1 \pmod 3, \\ -1 & \text{if } p \equiv 2 \pmod 3, \\ 0 & \text{if } p = 3, \end{cases}$$

which agrees on coprime arguments with the primitive Dirichlet character χ_3 mod 3 used in §5. The Dedekind zeta of $\mathbb{Q}(\sqrt{-3})$ then factors (by the conductor-discriminant theorem applied to the abelian extension $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$; see [Marcus, 1977, §3, Cor. to Thm. 20]) as

$$\boxed{\zeta_{\mathbb{Q}(\sqrt{-3})}(s) \;=\; \zeta(s) \cdot L(s, \chi_3).}$$

This factorization is **identical**, on the nose, to the 2-channel (ζ, L_{χ_3}) decomposition that Theorem 5.1 derived from the $\mathbb{Z}/6$ Hurwitz packet. The basis $\{\zeta, L_{\chi_3}\}$ of the 2D collapse is the basis of Galois eigenspaces of ζ_K under the $\mathbb{Z}/2$ Galois action.

Verification. The probe [`probe_DedekindFactorization_Qsqrt-3_ClaudeEinstein.py`] confirms the factorization by:

- Evaluating $\zeta(s) \cdot L(s, \chi_3)$ at $s = 2, 3, 1.5, 2 + 5i$ and comparing with a finite Euler-product approximation of $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \prod_{\mathfrak{p}}(1 - N\mathfrak{p}^{-s})^{-1}$ over prime ideals $\mathfrak{p}$ of $\mathbb{Z}[\zeta_6]$ of norm ≤ 200; agreement to 10^{-3} (the finite-Euler-product truncation error).
- The class number formula for the imaginary quadratic field $\mathbb{Q}(\sqrt{-3})$ (discriminant $D_K = -3$, class number $h_K = 1$, $w_K = 6$ roots of unity), gives

$$L(1, \chi_3) \;=\; \frac{2\pi\, h_K}{w_K \sqrt{|D_K|}} \;=\; \frac{2\pi \cdot 1}{6\sqrt{3}} \;=\; \frac{\pi}{3\sqrt{3}}.$$

Numerically the probe reports

$$L(1,\chi_3) \;\approx\; 0.6046000107, \qquad \frac{\pi}{3\sqrt{3}} \;\approx\; 0.6045997881,$$

agreement to within 3×10^{-7} — consistent with the probe's evaluation at $s = 1 + 10^{-6}$ (deliberately off the pole structure of $L(s, \chi_0^{(3)})$), where the residual offset $|L'(1,\chi_3)| \cdot 10^{-6}$ matches the observed discrepancy.

6.3 The Connes–Consani Picard groupoid (brief)

We summarize Connes & Consani [arXiv:2602.15941, Feb 2026] only insofar as needed to state the identification of §6.4. The reader is referred to their paper for the full development.

Connes and Consani interpret the adele class space of $\mathbb{Q}$ as the natural *monoidal extension* of the Picard group of $\overline{\mathrm{Spec}}\,\mathbb{Z}$. The objects are torsion-free rank-1 abelian groups L equipped with **rigidifying data** consisting of:

- a **norm** = Arakelov metric on the line bundle (Riemann sector data);
- a group morphism to $\mathbb{R}_{>0}$ (the adele-norm sector);
- a **combinatorial datum** parametrizing the roots of unity in the character dual of L.

The composition law is the tensor product of rank-1 groups (with their rigidifying structures). The **singular strata for spectral realization of L-functions** in this framework are the points where the zeta/L-functions have zeros and poles.

The framework is global; restricting to a single Galois cover gives a finite-cyclotomic specialization. The relevant special case for Paper I is the cover $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$.

6.4 The structural identification

Identification T3.3 (*Galois Identification with $\mathbb{Q}(\sqrt{-3})$; T3.3 of [RH-Done-Draft1]; HPo Addendum H, H1–H3). The 2-channel structure (ζ, L_{χ_3}) derived from the $\mathbb{Z}/6$ Hurwitz packet in §5 is, as a Galois module under the involution $K : r \to -r \pmod 6$, the Galois decomposition of the Connes–Consani Picard groupoid for the degree-2 cover $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$. Specifically:*

Connes–Consani datum	*2-channel Bures realization*
Rank-1 line bundle on $\overline{\mathrm{Spec}}\,\mathbb{Z}$	*ζ-channel (trivial extension)*
Rank-1 line bundle on $\overline{\mathrm{Spec}}\,\mathbb{Z}[\sqrt{-3}]$ *via Galois cover*	*2-channel pair* (ζ, L_{χ_3})
Arakelov metric (Riemann sector norm)	*Bures Hermitian metric* $h = (\lvert\Xi\rvert^2 + \lvert\mathcal{L}_3\rvert^2)^{-1}$
Combinatorial datum: parametrization of roots of unity in character dual	*T1.3's $\mathbb{Z}/6$ residue chart + K-involution*
Galois involution $\mathrm{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q})$	*K-involution* $K : r \to -r \pmod 6$
Singular strata for spectral realization of L	*Joint zeros of ζ and L_{χ_3}*
Tensor product of rank-1 groups = adele product	*Multiplicative product on* $\zeta \otimes L_{\chi_3}$

The Dedekind factorization $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \zeta(s) \cdot L(s,\chi_3)$ is the operator-theoretic manifestation of this Galois identification.

Justification of T3.3. The identification proceeds entry by entry, one row at a time.

Row 1 (rank-1 line bundle on $\overline{\mathrm{Spec}}\,\mathbb{Z}$). The base field $\mathbb{Q}$ has Dedekind zeta $\zeta_{\mathbb{Q}}(s) = \zeta(s)$; in the Connes–Consani framework this is the *trivial* rank-1 line bundle on $\overline{\mathrm{Spec}}\,\mathbb{Z}$, identified with the K-fixed ζ-channel of Theorem 5.2.

Row 2 (rank-1 line bundle on $\overline{\mathrm{Spec}}\,\mathbb{Z}[\sqrt{-3}]$). $\zeta_{\mathbb{Q}(\sqrt{-3})}(s)$ is the Dedekind zeta of the Galois cover. The factorization $\zeta_{\mathbb{Q}(\sqrt{-3})} = \zeta \cdot L_{\chi_3}$ of §6.2 expresses it as the product of the trivial line bundle with the L-function of the cover's primitive non-trivial character. The 2-channel pair (ζ, L_{χ_3}) is precisely the Connes–Consani data of the cover, read in the Galois eigenbasis.

Row 3 (Arakelov metric $\leftrightarrow$ Bures Hermitian metric). On a rank-1 line bundle the Arakelov norm is a Hermitian metric. For the Galois-cover line bundle viewed as $\mathbb{C}^2$ (Galois eigenbasis), this Hermitian metric is the Fubini–Study/Bures form $h = (|\Xi|^2 + |\mathcal{L}_3|^2)^{-1}$ on the 2-channel pair, with Ξ the K-fixed amplitude and $\mathcal{L}_3$ the K-anti-fixed amplitude; the explicit Bures-side construction is developed in [HPo-notes, Addendum C1].

Rows 4–5 (combinatorial datum + Galois involution). The roots of unity in $\mathbb{Q}(\sqrt{-3})$ are $\mu_6 = \{\zeta_6^k\}_{k=0}^5$, indexed by $\mathbb{Z}/6$. Theorem 5.2 identified the K-involution $K : r \to -r \pmod 6$ on the Hurwitz packet — equivalently, on $\mathbb{Z}/6$ via the DFT pairing — as the eigendecomposition into K-fixed (ζ) and K-anti-fixed (L_{χ_3}) channels. The Galois involution σ of $\mathrm{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q})$ acts on μ_6 as $\zeta_6^k \mapsto \zeta_6^{-k}$, which under the DFT pairing is exactly K. So K-involution = Galois involution, on the nose.

Row 6 (singular strata). The zeros of $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \zeta(s)L(s,\chi_3)$ are the (disjoint) union of the zeros of ζ and of L_{χ_3}, sorted by their Galois parity: K-fixed = ζ-zeros (parity $+1$); K-anti-fixed = L_{χ_3}-zeros (parity -1).

Row 7 (tensor product = adele product). The Galois cover's $\zeta_K = \zeta \cdot L_{\chi_3}$ is the multiplicative product corresponding to the tensor product of the underlying rank-1 line bundles (Connes–Consani composition law). This is the Dedekind factorization read at the L-function level. □

The identification is **structural** — it does not assert a new analytic theorem about ζ. What it does is name the program's 2-channel Bures fiber, in finite-cyclotomic coordinates on $\mathbb{Z}/6$, as the finite-arithmetic realization of the Connes–Consani Picard groupoid for the $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$ Galois cover.

6.5 Closing identification I3.4

We close §6 with the *closing identification* that ties Paper I's two main threads — the cubic effective action of §§2–4 and the Galois decomposition of §§5–6 — into one statement to be carried into Paper II.

Identification I3.4 (*The program's $\mathbb{Z}/6$ Bures carrier as the degree-2 cover of the Connes–Consani adele class space; T-handoff to Paper II*). *The program's $\mathbb{Z}/6$ Bures carrier (the 2-channel object derived in §5 from the Hurwitz packet collapse, with K-involution acting as the parity operator) is structurally the degree-2 cover by $\mathbb{Q}(\sqrt{-3})$ of the Connes–Consani adele class space of $\mathbb{Q}$.*

The cubic effective action $\Phi_\zeta(s) = \log\zeta(s)$ of Theorem 2.1, read on this 2-channel cover, decomposes under the Galois K-involution as $\Phi_\zeta = \log\zeta_K - \log L_{\chi_3}$, where $\zeta_K(s) = \zeta(s)L(s,\chi_3)$ is the Dedekind zeta of the cover. The Galois-even component is $\log\zeta$; the Galois-odd component is $\log L_{\chi_3}$, which Paper II identifies (after the Bures-covariant gauge subtraction of [HPo-notes Addendum B]) as the

connection — the gauge — under which the spectral-coordinate log-derivative $m_B = -\Xi'/\Xi$ on $\mathbb{C}_+$ becomes the natural Herglotz candidate for the CM Poisson Monotonicity criterion of Paper II.

This is the structural setup that Paper II requires. The 2-channel Bures fiber is the finite-arithmetic substrate; Paper II builds the right Hilbert space (the occupation-decorated symmetric Fock $\mathcal{F}_n$ at each primorial row, lifted to the Archimedean place via Stone–von Neumann uniqueness) on this substrate.

6.6 What I3.4 is, and is not

I3.4 is: - A structural identification of the program's $\mathbb{Z}/6$ Bures fiber with the Galois decomposition of $\zeta_{\mathbb{Q}(\sqrt{-3})}$; - The conceptual bridge from Paper I's finite-cyclotomic analysis to Paper II's representation-theoretic carrier; - The recognition that the K-involution of the program is the Galois involution of an actual number-theoretic Galois cover.

I3.4 is not: - An analytic theorem about ζ (no new analytic statement is asserted by I3.4 itself); - A construction of the Hilbert-Polya / de Branges operator (this is Paper II's terrain); - A proof of RH (this is Paper III).

The remaining Paper-I-internal task is to summarize what has been established and to hand off to Paper II.

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§7. Conclusion and Hand-off to Paper II

7.1 What Paper I established

Paper I has executed three classes of identifications and one closing structural claim.

Identification class 1 — the natural object on the trilogy's chart. The prime-axis exponential family on $\mathbb{N}_{>0}$ (Definition 2.1) is the canonical multi-index exponential family on which the trilogy's cumulant tower naturally lives. Its log-partition is $\log\zeta(s)$ (Theorem 2.1 = T2.1). The Euler product is the partition-function fingerprint of multiplicative independence of prime-axis sufficient statistics, and the Dirichlet series is the same partition function in dual mean coordinates. ζ is therefore not an analytic mystery imported from $\sum n^{-s}$; it is the categorical effective action $\Phi = \log Z$ of 2 , specialized to the prime-axis simplex along the slice $\eta_p(s) = -s\log p$.

Identification class 2 — the cubic structure in the residue chart. The Amari–Chentsov tensor of $\log\zeta$, read in T1.3's character basis on $\mathbb{Z}/N$ at the residue-projected zeta distribution, has the closed-form formula of Theorem 3.1 (= T2.2) — a polynomial in the additive-twist zetas $\zeta_N(s,k)$ with conductor structure inherited from the divisor lattice of N. This formula recovers the trilogy selection rule as its $s \to 1^+$ boundary value. Gauss-sum duality (§3.6) identifies the additive selection rule $a + b + c \equiv 0 \pmod N$ with the classical multiplicative-character orthogonality of triple L-function moments. The cubic functional equation (Theorem 4.1 = T2.3) pins the critical line $\mathrm{Re}(s) = 1/2$ as the unique locus along the primorial ladder on which the twisted Plancherel coefficients are N-uniformly bounded.

Identification class 3 — the 2D collapse and the Galois identification. At $N = 6$, the five non-trivial Hurwitz channels $\zeta(s, r/6)$ live in a 2-dimensional function space spanned by $\{\zeta(s), L(s,\chi_3)\}$ (Theorem 5.1 = T3.1), with closed-form coefficients. The K-involution $r \to -r$

(mod 6) block-diagonalizes the collapse into K-fixed ζ-channel and K-anti-fixed L_{χ_3}-channel (Theorem 5.2 = T3.2). The 2-channel structure is precisely the Galois decomposition of the Dedekind zeta of $\mathbb{Q}(\sqrt{-3})$ under the degree-2 cover, with K-involution = Galois involution and Dedekind factorization $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \zeta(s) \cdot L(s, \chi_3)$ (Identification 6.4 = T3.3).

Closing structural identification. The program's $\mathbb{Z}/6$ Bures carrier is the degree-2 cover by $\mathbb{Q}(\sqrt{-3})$ of the Connes–Consani adele class space (Identification 6.5 = I3.4).

Numerical verifications of the new closed forms (T3.1 and T3.3) are recorded in `probe_Hurwitz_z6_closed_form.p` (~10^{-60} analytical agreement) and `probe_DedekindFactorization_Qsqrt-3_ClaudeEinstein.py` (seven-digit agreement with the class number formula).

7.2 What Paper I does not establish

Paper I is the *identification phase* of the program. It establishes **what we are working with** — the natural object on the trilogy's chart and its Galois decomposition for the smallest squarefree multi-prime modulus. It does not yet:

(a) **Build the right Hilbert space.** The occupation-decorated symmetric Fock $\mathcal{F}_n = \mathrm{Sym}(\bigoplus_{p \le p_n} \mathbb{C}\, e_p)$ that hosts the exact von Mangoldt formula upstairs of the CRT compression is a Paper II object. Paper I does not construct it.

(b) **Promote the finite carrier to the Archimedean place.** The Uniform Place Treatment that extends $\mathcal{F}_n$ to $v = \infty$, with Stone–von Neumann uniqueness forcing the Bargmann–Fock $= L^2(\mathbb{R}, dx)$ realization with Tate-Weil normalization giving the standard $\pi^{-s/2}\Gamma(s/2)$ Archimedean local L-factors, is a Paper II construction.

(c) **State or use the CM Poisson Monotonicity criterion.** The characterization RH $\iff$ $\mathrm{Im}\, m_B(z) \ge 0$ on $\mathbb{C}_+$, with $m_B = -\Xi'/\Xi$ the Bures-covariant log-derivative of the spectral-coordinate completed zeta, is a Paper II statement (with the Bures-covariant K-gauge subtraction of [HPo-notes Addendum B] supplying the program's natural derivation).

(d) **Prove convolution-square positivity of Poisson windows, transport, or RH.** All of Paper III.

The boundary between Paper I and Paper II is the boundary between *what the program is working with* and *the carrier on which the analytic positivity machinery lives.* Paper I stays strictly on the first side.

7.3 Hand-off to Paper II

Paper II inherits from Paper I:

(i) **Theorem 2.1.** $\log \zeta$ is the categorical effective action on the prime-axis exponential family.

(ii) **Theorem 3.1.** Closed-form cubic A-C tensor on $\mathbb{Z}/N$ at the zeta distribution, as a polynomial in $\zeta_N(s, k)$.

(iii) **Theorem 4.1.** Critical-line characterization as the N-uniformity locus of the twisted Plancherel coefficients along the primorial ladder.

(iv) **Theorem 5.1, Theorem 5.2.** Closed-form $\zeta(s, r/6)$ in the (ζ, L_{χ_3}) basis, with K-involution as parity decomposition.

(v) **Identification T3.3 + I3.4.** The 2-channel Bures fiber as the finite-cyclotomic realization of the Connes–Consani Picard groupoid for $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$; K-involution = Galois involution; Dedekind factorization $\zeta_K = \zeta \cdot L_{\chi_3}$.

(vi) **Theorem 5.3.** $L(1/2, \chi_3) > 0$ unconditionally (Iwaniec–Sarnak

- LMFDB).

What Paper II must contribute beyond Paper I:

- **The occupation-decorated symmetric Fock** $\mathcal{F}_n$ at each primorial row n, with the *single-axis projector* Π_{sa} that distinguishes mixed-axis occupation histories from single-axis ones before CRT compression. The exact upstairs von Mangoldt formula $\text{VM}_{n,K}(G) = D_s \, \Pi_{\text{sa}} \sum_k k^{-1} H_s^k$ with $H_s = \sum_p p^{-s} e_p$.
- **Uniform Place Treatment** including the Archimedean local factor $\mathcal{F}_\infty = \text{Sym}(\mathbb{C}\, e_\infty) = \mathbb{C}[e_\infty]$, with a Heisenberg algebra realization $[\hat{a}, \hat{a}^\dagger] = 1$ on this factor. Stone–von Neumann uniqueness then forces the Hilbert-space completion to be Bargmann–Fock $= L^2(\mathbb{R}, dx)$, with Tate-Weil normalization giving the standard Archimedean local L-factors $\pi^{-s/2}\Gamma(s/2)$ (Tate's local zeta integral on the ground state) and $\pi^{-(s+1)/2}\Gamma((s+1)/2)$ (first excited state).
- **The Bures-covariant log-derivative** $m_B = -\Xi'/\Xi$ identified (via the K-odd gauge subtraction of [HPo-notes Addendum B]) on the spectral coordinate z with $s = 1/2 + iz$. CM Poisson Monotonicity: $\text{Im}\, m_B(z) \geq 0$ on $\mathbb{C}_+ \iff$ RH (Polya–Laguerre, classical).

Paper I has identified *what* the program is working with on $\mathbb{Z}/6$. Paper II identifies *the carrier* on which m_B lives, equipping it with the right finite arithmetic at each primorial row and the right Archimedean factor at $v = \infty$.

7.4 The arc, in one paragraph

The trilogy `Identifying 3` gave us the chart, the cumulant tower, and the character basis on the conductor lattice. Paper I (the present paper) names $\log \zeta$ as the natural object on that chart, derives its cubic structure in closed form, pins the critical line as a structural locus on the primorial ladder, and identifies the smallest squarefree multi-prime modulus's Hurwitz packet with the Galois decomposition of $\zeta_{\mathbb{Q}(\sqrt{-3})}$. Paper II will build the right Hilbert carrier on this substrate — occupation-decorated symmetric Fock at each primorial row, Uniform Place Treatment extending to the Archimedean factor, Bures-covariant log-derivative as the Hilbert-Polya/CM Poisson target. Paper III will close to RH via convolution-square positivity of Poisson windows, averaged CRT shadow uniformity at polynomial scale, and positive-current transport. Paper I establishes the **finite arithmetic substrate**; the analytic positivity machinery lives elsewhere.

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`probe_Hurwitz_z6_closed_form.py` — verifies Theorem 5.1 (closed-form $\zeta(s, r/6)$) at 60-digit precision; verifies Theorem 5.2 (K-even vector identities) at zeta-zero ordinates to machine precision.

`probe_DedekindFactorization_Qsqrt-3_ClaudeEinstein.py` — verifies Identification T3.3 (Dedekind factorization $\zeta_K = \zeta \cdot L_{\chi_3}$) against finite Euler product over prime ideals of norm ≤ 200;

verifies class number formula $L(1, \chi_3) = \pi/(3\sqrt{3})$ to seven digits.